



SAPIENZA
UNIVERSITÀ DI ROMA

Modellazione

MIP

Numerical problems

Fifteen numerical models, from EX 1 to EX 15: explicit data, few variables, one step for each technique. They are the easiest examples of the course and are read to get the measure of things, before the general problems with a symbolic model. For each one: statement, model, instance, a feasible solution, the dual of the relaxation and the optimum, with the script that reproduces them.

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Online version and downloads:

fabiofurini.github.io/modellazione-mip/
github.com/fabiofurini/modellazione-mip

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Fifteen numerical models

These notes collect the fifteen numerical models of the course, from EX 1 to EX 15. Explicit data, few variables, one technique per model: they are the easiest ones, and they come before the three families of problems and the mixed problems.

Every numerical model always has the same five parts:

- the statement, with the data of the instance;
- the **variables**, with their domain and their count;
- the **model of the instance**, primal and dual, in the violet box;
- a feasible solution built by hand, which gives the primal bound;
- a dual solution built by hand, which gives the dual bound, and the comparison with the solver optimum.

Every model has its own script and its own notebook, one per model, listed at the beginning of each section.

EX 1 — The eight-seat van

Script: `python/ex01_van.py`

[Notebook in Colab](#)

EX 1 — which groups of tourists to accept

A tour operator rents an eight-seat van to guide tourists around a scenic location. On a given day four groups show up, of 2, 3, 4 and 5 people. Since the van cannot take all of them at once, the operator asks each group (whose members do not want to be separated) to make an offer: the four groups offer 30, 50, 80 and 70 euros. For operational reasons at most two groups can be accepted; moreover, if group 2 is accepted then group 4 must be accepted as well. The goal is to maximise the total earnings.

The model.

With $x_j = 1$ if group j is accepted:

EX 1 on the instance — primal

$$\begin{aligned}
& \max 30x_1 + 50x_2 + 80x_3 + 70x_4 \\
& \text{subject to } 2x_1 + 3x_2 + 4x_3 + 5x_4 \leq 8 \\
& \quad x_1 + x_2 + x_3 + x_4 \leq 2 \\
& \quad \quad x_2 - x_4 \leq 0 \\
& \quad x_1, \quad x_2, \quad x_3, \quad x_4 \in \{0, 1\}
\end{aligned}$$

Description. Three techniques of chapter “Links between variables” in four lines: the capacity (section 3.1), the count (3.4) and the logical precedence (3.9). The third constraint reads $x_2 \leq x_4$: if $x_2 = 1$ then $x_4 = 1$. The other direction is false and must not hold: group 4 can be accepted without group 2.

Heuristic: the primal bound.

Constructive heuristic on the decreasing offer, accepting a group only if all the constraints stay satisfied (the implication included).

- Group 3 (offer 80): accepted, 4 seats, 1 group.
- Group 4 (offer 70): rejected, 9 seats would be needed out of 8.
- Group 2 (offer 50): rejected, it would force accepting group 4.
- Group 1 (offer 30): accepted, 6 seats, 2 groups.

$$z(MILP) \geq LB = 110.$$

Dual: the dual bound.

With $\alpha \geq 0$ on the seats, $\beta \geq 0$ on the number of groups and $\gamma \geq 0$ on the implication, the dual is

$$\min 8\alpha + 2\beta \quad \text{s.t.} \quad a_j \alpha + \beta (+\gamma \text{ if } j = 2, -\gamma \text{ if } j = 4) \geq p_j.$$

The recipe: $\beta = \gamma = 0$ and only the seat is priced, at the price no group can beat,

$$\bar{\alpha} = \max_j \frac{p_j}{a_j} = \max\left(\frac{30}{2}, \frac{50}{3}, \frac{80}{4}, \frac{70}{5}\right) = 20, \quad UB = 8 \cdot 20 = 160.$$

EX 1 on the instance — dual of the LP relaxation

$$\begin{aligned}
& \min 8\alpha + 2\beta \\
& \text{subject to } 2\alpha + \beta \geq 30 \\
& \quad 3\alpha + \beta + \gamma \geq 50 \\
& \quad 4\alpha + \beta \geq 80 \\
& \quad 5\alpha + \beta - \gamma \geq 70 \\
& \quad \alpha \geq 0 \\
& \quad \beta \geq 0 \\
& \quad \gamma \geq 0
\end{aligned}$$

Results.

	LB	UB	$z(LP)$	$z(LP^+)$	$z(MILP)$
EX 1 van (max)	110	160	160	140	120

The optimum accepts groups 2 and 4, filling exactly the eight seats, for earnings of 120. The hand-built dual bound is *optimal* for the relaxation without the bounds (160); the strengthened one, which adds $x_j \leq 1$, drops to 140 and gets closer to the optimum. Removing the implication the optimum would rise to 130 (groups 2 and 3): the implication costs 10 euros. Removing the limit of two groups changes nothing, because it is the seat capacity that bites.

EX 2 — Bus lines

Script: `python/ex02_buslines.py`[Notebook in Colab](#)

EX 2 — assignment with a capacity in number of lines

A city council is evaluating the offers of three bus companies for four bus lines. Each company has stated the cost at which it would operate each line:

Company	Line 1	Line 2	Line 3	Line 4
1	10	4	9	7
2	1	2	3	10
3	8	9	10	1

Every line must be assigned to exactly one company and every company may operate at most two lines. The total cost is to be minimised.

The model.

With $x_{ij} = 1$ if company i operates line j ($3 \cdot 4 = 12$ binary variables):

EX 2

$$\begin{aligned}
 & \min \sum_{i=1}^n \sum_{j=1}^m c_{ij} x_{ij} \\
 & \text{subject to} \quad \sum_{i=1}^n x_{ij} = 1, & \forall j \in \{1, 2, \dots, m\}, \\
 & \quad \sum_{j=1}^m x_{ij} \leq p, & \forall i \in \{1, 2, \dots, n\}, \\
 & \quad x_{ij} \in \{0, 1\}, & \forall i \in \{1, 2, \dots, n\}, \forall j \in \{1, 2, \dots, m\}.
 \end{aligned}$$

Description. With $n = 3$ companies, $m = 4$ lines and $p = 2$: four *set partitioning* constraints (one per line) and three capacity constraints (one per company). It is problem 7.1 with the capacity counted in *number* of jobs instead of time.

Heuristic.

Constructive heuristic on the lines, in order: every line goes to the cheapest company among those not yet full.

- **Line 1:** costs 10, 1, 8; the cheapest is company 2.
- **Line 2:** costs 4, 2, 9; company 2 again, which is now full.
- **Line 3:** companies 1 and 3 remain, costs 9 and 10: company 1.
- **Line 4:** companies 1 and 3 remain, costs 7 and 1: company 3.

Value $1 + 2 + 9 + 1 = 13$, so $UB = 13$.

Dual and lower bound.

Relaxing $x_{ij} \in \{0, 1\}$ to $x_{ij} \geq 0$, with α_j free for every line constraint and $\pi_i \leq 0$ for every capacity constraint:

Dual of the LP relaxation of EX 2

$$\begin{aligned}
 \max \quad & \sum_{j=1}^m \alpha_j + p \sum_{i=1}^n \pi_i \\
 \text{subject to} \quad & \alpha_j + \pi_i \leq c_{ij}, & \forall i \in \{1, 2, \dots, n\}, \forall j \in \{1, 2, \dots, m\}, \\
 & \alpha_j \geq 0, & \forall j \in \{1, 2, \dots, m\}, \\
 & \pi_i \leq 0, & \forall i \in \{1, 2, \dots, n\}.
 \end{aligned}$$

Description. The variable α_j is the value of line j , which has to be assigned anyway, and π_i the (non-positive) price of one of the p lines company i may run. The objective prices the lines and the available slots. The dual constraints are the columns of the x_{ij} : giving line j to company i satisfies its assignment constraint and takes one slot, and the balance cannot exceed the cost c_{ij} of that pairing.

With the "zero out and saturate" recipe, $\bar{\pi} = 0$ and $\bar{\alpha}_j = \min_i c_{ij} = (1, 2, 3, 1)$, feasible by construction: $LB = 7$. Meaning: "every line costs at least the lowest offer it received".

EX 2 on the instance — dual of the LP relaxation

$$\begin{aligned}
& \max \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 + 2\beta_1 + 2\beta_2 + 2\beta_3 \\
& \text{subject to } \alpha_1 + \beta_1 \leq 10 \\
& \quad \alpha_2 + \beta_1 \leq 4 \\
& \quad \alpha_3 + \beta_1 \leq 9 \\
& \quad \alpha_4 + \beta_1 \leq 7 \\
& \quad \alpha_1 + \beta_2 \leq 1 \\
& \quad \alpha_2 + \beta_2 \leq 2 \\
& \quad \alpha_3 + \beta_2 \leq 3 \\
& \quad \alpha_4 + \beta_2 \leq 10 \\
& \quad \alpha_1 + \beta_3 \leq 8 \\
& \quad \alpha_2 + \beta_3 \leq 9 \\
& \quad \alpha_3 + \beta_3 \leq 10 \\
& \quad \alpha_4 + \beta_3 \leq 1 \\
& \alpha_1, \alpha_2, \alpha_3, \alpha_4 \geq 0 \\
& \beta_1, \beta_2, \beta_3 \geq 0
\end{aligned}$$

The bounds.

The optimum is $z(MILP) = 9$ (lines 1 and 3 to company 2, line 2 to company 1, line 4 to company 3) and both relaxations coincide with it.

UB (constructive heuristic)	LB (dual by hand)	$z(LP)$	$z(MILP)$	heuristic gap
13	7	9	9	44.4%

Worth noting: $z(LP) = z(MILP) = 9$, yet the hand-built bound stops at 7 and the heuristic is off by 44%. The relaxation is exact, the dual recipe is not: they are two different things.

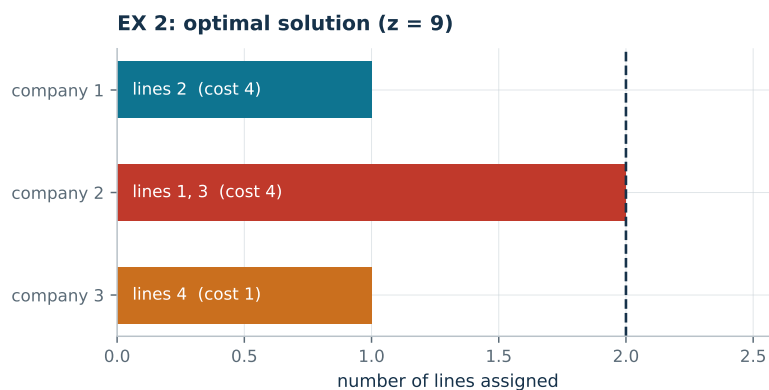


Figure 1: EX 2: the optimal solution. No company exceeds two lines.

EX 3 — Relay

Script: `python/ex03_relay.py`

[Notebook in Colab](#)

EX 3 — assignment with more resources than tasks

A coach must form a relay team by choosing three swimmers out of four. The race has three 50-metre legs: backstroke, breaststroke and butterfly. The recorded times (seconds) are:

Stroke	Carlo	Paolo	Fabio	Marco
Backstroke	37	32	33	37
Breaststroke	43	33	42	34
Butterfly	33	28	38	30

The total relay time is to be minimised.

The model.

With $x_{sn} = 1$ if stroke s is swum by n : $\sum_n x_{sn} = 1$ for every stroke (3 constraints), $\sum_s x_{sn} \leq 1$ for every swimmer (4 constraints), objective $\min \sum_{s,n} t_{sn} x_{sn}$. One swimmer stays out: it is an assignment with more resources than tasks.

EX 3 on the instance — primal

$$\begin{aligned}
 & \min 37x_{11} + 32x_{12} + 33x_{13} + 37x_{14} + 43x_{21} + 33x_{22} + 42x_{23} + 34x_{24} + 33x_{31} + 28x_{32} + 38x_{33} + 30x_{34} \\
 \text{subject to } & x_{11} + x_{12} + x_{13} + x_{14} = 1 \\
 & x_{21} + x_{22} + x_{23} + x_{24} = 1 \\
 & x_{31} + x_{32} + x_{33} + x_{34} = 1 \\
 & x_{11} + x_{21} + x_{31} \leq 1 \\
 & x_{12} + x_{22} + x_{32} \leq 1 \\
 & x_{13} + x_{23} + x_{33} \leq 1 \\
 & x_{14} + x_{24} + x_{34} \leq 1 \\
 & x_{11}, x_{12}, x_{13}, x_{14}, x_{21}, x_{22}, x_{23}, x_{24}, x_{31}, x_{32}, x_{33}, x_{34} \in \{0, 1\}
 \end{aligned}$$

Heuristic.

Constructive heuristic on the strokes in the given order: backstroke to Paolo (32), breaststroke to Marco (34, the fastest of the three left), butterfly to Carlo (33). Total $UB = 99$ seconds.

Dual and lower bound.

The same structure as EX 2 with $p = 1$: $\bar{\pi} = 0$ and $\bar{\alpha}_s = \min_n t_{sn} = (32, 33, 28)$, so $LB = 93$. Meaning: "every stroke costs at least the time of its best specialist". The bound is weak exactly when two strokes share the same best specialist — here Paolo is the fastest in both backstroke and butterfly, and indeed the bound is not attainable.

EX 3 on the instance — dual of the LP relaxation

$$\begin{aligned}
& \max \alpha_1 + \alpha_2 + \alpha_3 + \beta_1 + \beta_2 + \beta_3 + \beta_4 \\
& \text{subject to } \alpha_1 + \beta_1 \leq 37 \\
& \quad \alpha_1 + \beta_2 \leq 32 \\
& \quad \alpha_1 + \beta_3 \leq 33 \\
& \quad \alpha_1 + \beta_4 \leq 37 \\
& \quad \alpha_2 + \beta_1 \leq 43 \\
& \quad \alpha_2 + \beta_2 \leq 33 \\
& \quad \alpha_2 + \beta_3 \leq 42 \\
& \quad \alpha_2 + \beta_4 \leq 34 \\
& \quad \alpha_3 + \beta_1 \leq 33 \\
& \quad \alpha_3 + \beta_2 \leq 28 \\
& \quad \alpha_3 + \beta_3 \leq 38 \\
& \quad \alpha_3 + \beta_4 \leq 30 \\
& \alpha_1, \alpha_2, \alpha_3 \geq 0 \\
& \beta_1, \beta_2, \beta_3, \beta_4 \geq 0
\end{aligned}$$

The bounds.

The optimum is $z(MILP) = 95$ seconds: backstroke to Fabio (33), breaststroke to Marco (34), butterfly to Paolo (28); Carlo stays out.

UB (constructive heuristic)	LB (dual by hand)	$z(LP)$	$z(MILP)$	heuristic gap
99	93	95	95	4.2%

Why the relaxation is exact here

The matrix of the model is that of the assignment problem: totally unimodular. All the vertices of the relaxation are integer, so $z(LP) = z(MILP)$ and integrality costs nothing — the same phenomenon as alldiff in Section “Alldiff and binary expansion”. This stops being true as soon as a capacity constraint with coefficients other than 0 and 1 is added.

EX 4 — Shoes: production, inventory and hirings

Script: `python/ex04_shoes.py`

[Notebook in Colab](#)

EX 4 — production and workforce over three months

Over the next three months a shoe factory must meet monthly demands of 3000, 5000 and 7000 pairs of shoes. At the start of the first month it has 500 pairs in stock and employs 100 workers. Every worker is paid 1500 euros a month and works 160 hours; producing one pair requires 4 labour hours and 15 euros of raw materials. At the start of every month workers may be hired, at a cost of 100 euros each for training. Storing one unsold pair at the end of the month costs 3 euros. The factory wants the production and workforce plan of minimum total cost.

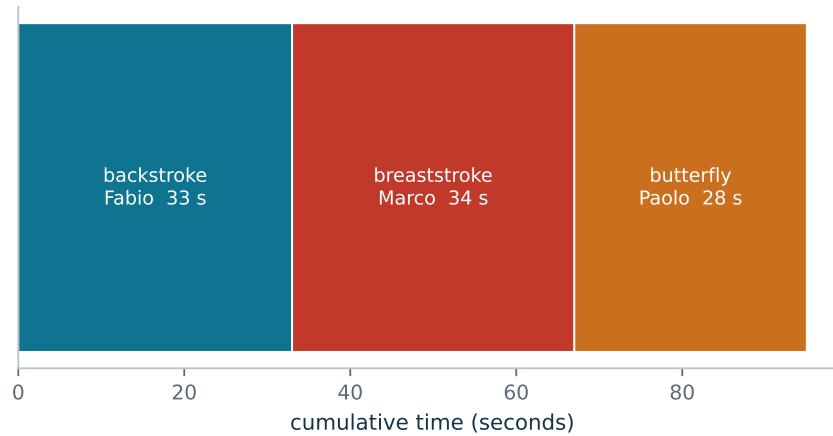
EX 3: the optimal relay (z = 95 s); Carlo stays out

Figure 2: EX 3: the optimal relay, leg by leg.

The model.

It is model 9.2A with one more datum, the initial inventory. With x_t pairs produced, s_t inventory at the end of the month, y_t workers on duty and z_t workers hired:

EX 4 on the instance

$$\begin{aligned}
 &\min 15x_1 + 15x_2 + 15x_3 + 3s_1 + 3s_2 + 1500y_1 + 1500y_2 + 1500y_3 + 100z_1 + 100z_2 + 100z_3 \\
 \text{subject to } &x_1 - s_1 = 2500 \\
 &x_2 + s_1 - s_2 = 5000 \\
 &x_3 + s_2 = 7000 \\
 &-4x_1 + 160y_1 \geq 0 \\
 &-4x_2 + 160y_2 \geq 0 \\
 &-4x_3 + 160y_3 \geq 0 \\
 &y_1 - z_1 = 100 \\
 &-y_1 + y_2 - z_2 = 0 \\
 &-y_2 + y_3 - z_3 = 0 \\
 &x_1, x_2, x_3 \geq 0 \\
 &s_1, s_2 \geq 0 \\
 &y_1, y_2, y_3 \in \mathbb{Z}_{\geq 0} \\
 &z_1, z_2, z_3 \in \mathbb{Z}_{\geq 0}
 \end{aligned}$$

Description. The demand of the first month is $3000 - 500 = 2500$ net pairs; 14 500 pairs must be produced in total. Here the workforce appears in two families of variables at once, y_t and z_t , tied by the staff balance: it is the mixed formulation one obtains by keeping both readings of problem 9.2.

Heuristic: the primal bound.

"Just in time" production: every month the net demand is produced, with no inventory, hiring as many workers as needed. Month 1: 10 000 hours, 63 workers, the 100 already on duty are

enough. Month 2: 20 000 hours, 125 workers, 25 are hired. Month 3: 28 000 hours, 175 workers, another 50 are hired. The cost is

$$z(MILP) \leq UB = 825\,000.$$

Dual: the dual bound.

With α_t free on the demand balances, $\beta_t \geq 0$ on the hours and γ_t free on the workforce balances, the recipe is: $\gamma = 0$, and one hour of work is worth what it really costs,

$$\bar{\beta}_t = \frac{w}{r} = \frac{1500}{160} = \frac{75}{8},$$

so that one pair is worth at most

$$\bar{\alpha}_t = 15 + 4 \cdot \frac{75}{8} = \frac{105}{2} = 52.5.$$

One checks that all the dual columns are satisfied, and

$$LB = \frac{105}{2} \cdot 14\,500 = 761\,250.$$

Dual of EX 4 on the instance

$$\begin{array}{llllll} \max & 2500\alpha_1 + 5000\alpha_2 + 7000\alpha_3 & & +100\gamma_1 & & \\ \text{subject to} & \alpha_1 & & -4\beta_1 & & \leq 15 \\ & & \alpha_2 & & -4\beta_2 & \leq 15 \\ & & & \alpha_3 & & -4\beta_3 \leq 15 \\ & -\alpha_1 & +\alpha_2 & & & \leq 3 \\ & & -\alpha_2 & +\alpha_3 & & \leq 3 \\ & & & 160\beta_1 & +\gamma_1 - \gamma_2 & \leq 1500 \\ & & & & 160\beta_2 & +\gamma_2 - \gamma_3 \leq 1500 \\ & & & & & 160\beta_3 & +\gamma_3 \leq 1500 \\ & & & & & & -\gamma_1 \leq 100 \\ & & & & & & -\gamma_2 \leq 100 \\ & & & & & & -\gamma_3 \leq 100 \\ & \alpha_1, & \alpha_2, & \alpha_3 & & & \geq 0 \\ & & & & \beta_1, & \beta_2, & \beta_3 & \geq 0 \\ & & & & & & & \gamma_1, \gamma_2, \gamma_3 \geq 0 \end{array}$$

Results.

	UB	LB	$z(LP)$	$z(LP^+)$	$z(MILP)$
EX 4 shoes	825 000	761 250	773 500	773 500	774 180

The optimal plan produces 3980, 4000 and 6520 pairs with 100, 100 and 163 workers: 1480 pairs are kept in stock at the end of the first month and 480 at the end of the second, and 63 workers are hired once only, in the third month. Producing early costs 3 euros a pair and saves wages: raising the storage cost to 20 euros the inventories drop to 1000 and 0 (optimum 807 500), at 60 euros they disappear altogether and one falls back on the heuristic (825 000).

EX 5 — Vehicles with a minimum quantity

Script: `python/ex05_vehicles.py`

[Notebook in Colab](#)

EX 5 — five types of vehicle

A company is considering the production of five vehicle types: compact car, midsize car, large car, midsize minivan and large minivan. It has 1000 tons of steel and 2000 labour hours available. The table reports, for every type, the steel per unit, the hours per unit, the unit profit and the minimum quantity to be produced if the type is selected:

Type	steel	hours	profit	minimum
compact car	2	30	200	10
midsize car	3	25	250	10
large car	5	40	300	10
midsize minivan	6	45	550	5
large minivan	8	55	700	5

The company wants the plan of maximum profit.

The model.

It is model 9.3 without the bonus: the same semicontinuous variables, with $M_j = \min_i \lfloor b_i / a_{ij} \rfloor$. On the instance $M = (66, 80, 50, 44, 36)$, and in every case the limiting resource is the labour time.

EX 5 on the instance — primal

$$\begin{array}{llllll}
 \max & 200x_1 + 250x_2 + 300x_3 + 550x_4 + 700x_5 \\
 \text{subject to} & 2x_1 + 3x_2 + 5x_3 + 6x_4 + 8x_5 & & & & \leq 1000 \\
 & 30x_1 + 25x_2 + 40x_3 + 45x_4 + 55x_5 & & & & \leq 2000 \\
 & x_1 & & & -10y_1 & \geq 0 \\
 & & x_2 & & -10y_2 & \geq 0 \\
 & & & x_3 & & -10y_3 \geq 0 \\
 & & & & x_4 & -5y_4 \geq 0 \\
 & & & & & x_5 - 5y_5 \geq 0 \\
 & x_1 & & & -66y_1 & \leq 0 \\
 & & x_2 & & -80y_2 & \leq 0 \\
 & & & x_3 & & -50y_3 \leq 0 \\
 & & & & x_4 & -44y_4 \leq 0 \\
 & & & & & x_5 - 36y_5 \leq 0 \\
 & x_1, & x_2, & x_3, & x_4, & x_5 & \in \mathbb{Z}_{\geq 0} \\
 & & & & & & y_1, y_2, y_3, y_4, y_5 \in \{0, 1\}
 \end{array}$$

Heuristic: the primal bound.

Constructive heuristic on the profit per labour hour: 20/3, 10, 15/2, 110/9 and 140/11 for the five types. One starts from the large minivan (140/11), produces 36 of them (the maximum

allowed by the hours), and is left with 712 tons of steel but only 20 hours: no other type reaches its own minimum quantity and the constructive heuristic stops.

$$z(MILP) \geq LB = 25\,200.$$

Dual: the dual bound.

With $\ell_j = \beta_j = 0$ (the minimum lot is not priced) and a single resource priced at the highest unit price among the types:

$$\begin{aligned}\bar{\pi}_{\text{steel}} &= \max_j \frac{\bar{p}_j}{a_{1j}} = 100 \Rightarrow 1000 \cdot 100 = 100\,000, \\ \bar{\pi}_{\text{hours}} &= \max_j \frac{\bar{p}_j}{a_{2j}} = \frac{140}{11} \Rightarrow 2000 \cdot \frac{140}{11} = \frac{280\,000}{11} \approx 25\,454.5.\end{aligned}$$

The better bound comes from the hours:

$$z(MILP) \leq UB = \frac{280\,000}{11}.$$

EX 5 on the instance — dual

$$\begin{array}{llllllllllll} \min & 1000\pi_1 + 2000\pi_2 & & & & & & & & & & & \\ \text{subject to} & 2\pi_1 & +30\pi_2 & +\lambda_1 & & & & +\mu_1 & & & & & \geq 200 \\ & 3\pi_1 & +25\pi_2 & & +\lambda_2 & & & & +\mu_2 & & & & \geq 250 \\ & 5\pi_1 & +40\pi_2 & & & +\lambda_3 & & & & +\mu_3 & & & \geq 300 \\ & 6\pi_1 & +45\pi_2 & & & & +\lambda_4 & & & & +\mu_4 & & \geq 550 \\ & 8\pi_1 & +55\pi_2 & & & & & +\lambda_5 & & & & +\mu_5 & \geq 700 \\ & & & -10\lambda_1 & & & & & -66\mu_1 & & & & \geq 0 \\ & & & & -10\lambda_2 & & & & & -80\mu_2 & & & \geq 0 \\ & & & & & -10\lambda_3 & & & & & -50\mu_3 & & \geq 0 \\ & & & & & & -5\lambda_4 & & & & & -44\mu_4 & \geq 0 \\ & & & & & & & -5\lambda_5 & & & & -36\mu_5 & \geq 0 \\ & \pi_1, & \pi_2 & & & & & & & & & & \geq 0 \\ & & & \lambda_1, & \lambda_2, & \lambda_3, & \lambda_4, & \lambda_5 & & & & & \geq 0 \\ & & & & & & & & \mu_1, & \mu_2, & \mu_3, & \mu_4, & \mu_5 \geq 0 \end{array}$$

Results.

	LB	UB	$z(LP)$	$z(LP^+)$	$z(MILP)$
EX 5 vehicles (max)	25 200	280000/11	280000/11	229000/9	25 250

The optimum produces 9 midsize minivans and 29 large minivans, using all 2000 hours and 286 tons of steel out of 1000. The dual bound is *optimal* for the relaxation without the bounds, and the heuristic is off by a mere 0.2%. Without minimum quantities the optimum would be 25 300: the minimum lot costs 50 euros of profit. Doubling the minimum quantities one drops to 25 200, and doubling the labour hours one reaches 50 750 with the hours still saturated.

EX 6 — Hub-and-spoke

Script: `python/ex06_hub.py`

[Notebook in Colab](#)

EX 6 — the minimum number of hubs

An airline organises its network with a *hub-and-spoke* system: a hub serves directly every city within 1000 miles. The company operates on eight cities; for each of them, the cities within 1000 miles are:

City	Cities within 1000 miles
Atlanta	Atlanta, Chicago, Houston, New York
Chicago	Atlanta, Chicago, New York
Denver	Denver, Los Angeles
Houston	Atlanta, Houston
Los Angeles	Denver, Los Angeles, San Francisco
New York	Atlanta, Chicago, New York
San Francisco	Los Angeles, San Francisco, Seattle
Seattle	San Francisco, Seattle

The minimum number of hubs is wanted, such that every city is within 1000 miles of at least one hub.

The model.

The data are $n = 8$ cities and $m = 8$ candidate hubs. With $y_j = 1$ if city j becomes a hub (m binaries) and S_i the set of cities covering city i :

EX 6

$$\begin{aligned}
 & \min \sum_{j=1}^m y_j \\
 & \text{subject to } \sum_{j \in S_i} y_j \geq 1, & \forall i \in \{1, 2, \dots, n\}, \\
 & y_j \in \{0, 1\}, & \forall j \in \{1, 2, \dots, m\}.
 \end{aligned}$$

Description. A pure *set covering*: eight covering constraints, all costs equal to 1. Note that "within 1000 miles" is a symmetric relation, so $j \in S_i \iff i \in S_j$: the matrix of the model is symmetric.

EX 6 on the instance — primal

$$\begin{aligned}
& \min y_1 + y_2 + y_3 + y_4 + y_5 + y_6 + y_7 + y_8 \\
& \text{subject to } y_1 + y_2 + y_4 + y_6 \geq 1 \\
& \quad y_1 + y_2 + y_6 \geq 1 \\
& \quad y_3 + y_5 \geq 1 \\
& \quad y_1 + y_4 \geq 1 \\
& \quad y_3 + y_5 + y_7 \geq 1 \\
& \quad y_1 + y_2 + y_6 \geq 1 \\
& \quad y_5 + y_7 + y_8 \geq 1 \\
& \quad y_7 + y_8 \geq 1 \\
& \quad y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8 \in \{0, 1\}
\end{aligned}$$

Heuristic.

Constructive covering heuristic (Chapter “Constructive heuristics”), with the criterion "cost per newly covered city".

- **Step 1.** Ratios 1/4 for Atlanta, 1/3 for Chicago, New York and San Francisco, 1/2 for the others: **Atlanta** is chosen, covering Atlanta, Chicago, Houston and New York.
- **Step 2.** Denver, Los Angeles, San Francisco and Seattle remain; ratios 1/2 for Denver, 1/3 for Los Angeles and San Francisco, 1/2 for Seattle: **Los Angeles** is chosen, covering Denver, Los Angeles and San Francisco.
- **Step 3.** Seattle remains; ratio 1/1 for San Francisco and for Seattle: **San Francisco** is chosen.

Three hubs, so $UB = 3$.

Dual and lower bound.

Relaxing to $y_j \geq 0$, with $\pi_i \geq 0$ for every covering constraint ($\geq$ direction in a minimisation):

Dual of the LP relaxation of EX 6

$$\begin{aligned}
& \max \sum_{i=1}^n \pi_i \\
& \text{subject to } \sum_{i: j \in S_i} \pi_i \leq 1, & \forall j \in \{1, 2, \dots, m\}, \\
& \quad \pi_i \geq 0, & \forall i \in \{1, 2, \dots, n\}.
\end{aligned}$$

Description. The variable π_i is the price attached to city i . The objective adds up the prices of the eight cities. The constraints are the columns of the x_j , one per hub: the cities hub j covers, priced at those values, cannot be worth more than what it costs to open it, namely 1.

Dual constructive heuristic on the cities, in order: π_i is raised until the first opposing dual constraint becomes tight, and the residuals are updated.

- **Atlanta** is covered by four hubs, all with residual 1: $\bar{\pi}_1 = 1$, and the residuals of Atlanta, Chicago, Houston and New York drop to zero.
- **Chicago, Houston** and **New York** have all their hubs at residual zero: $\bar{\pi} = 0$.
- **Denver** is covered by Denver and Los Angeles, both with residual 1: $\bar{\pi}_3 = 1$, and their residuals drop to zero.
- **Los Angeles** and **San Francisco** have one hub at residual zero: $\bar{\pi} = 0$.
- **Seattle** is covered by San Francisco and Seattle, both with residual 1: $\bar{\pi}_8 = 1$.

Value $LB = 1 + 1 + 1 = 3$.

What this bound proves, in words

The three cities that received $\pi_i = 1$ — Atlanta, Denver, Seattle — are pairwise "far apart": no city can serve as a hub for two of them. Hence at least three hubs are needed. The dual has formalised an argument one could tell without writing it: that is the point of the constructive heuristic recipe.

EX 6 on the instance — dual of the LP relaxation

$$\begin{aligned}
 & \max \quad u_1 + u_2 + u_3 + u_4 + u_5 + u_6 + u_7 + u_8 \\
 & \text{subject to} \quad u_1 + u_2 \quad \quad + u_4 \quad \quad + u_6 \quad \leq 1 \\
 & \quad \quad \quad u_1 + u_2 \quad \quad \quad + u_6 \quad \leq 1 \\
 & \quad \quad \quad \quad \quad u_3 \quad \quad + u_5 \quad \leq 1 \\
 & \quad \quad \quad u_1 \quad \quad \quad + u_4 \quad \leq 1 \\
 & \quad \quad \quad \quad \quad u_3 \quad \quad + u_5 \quad + u_7 \leq 1 \\
 & \quad \quad \quad u_1 + u_2 \quad \quad \quad + u_6 \quad \leq 1 \\
 & \quad \quad \quad \quad \quad \quad \quad u_5 \quad \quad + u_7 + u_8 \leq 1 \\
 & \quad \quad \quad \quad \quad \quad \quad \quad \quad u_7 + u_8 \leq 1 \\
 & \quad \quad \quad u_1, \quad u_2, \quad u_3, \quad u_4, \quad u_5, \quad u_6, \quad u_7, \quad u_8 \geq 0
 \end{aligned}$$

The bounds.

$z(MILP) = 3$, with hubs in Atlanta, Los Angeles and San Francisco — the same solution as the constructive heuristic.

UB (constructive heuristic)	LB (dual by hand)	$z(LP)$	$z(MILP)$	heuristic gap
3	3	3	3	0.0%

The two hand-built bounds coincide: optimality is proved without the solver.

EX 7 — Custom aircraft with a fixed set-up cost

Script: `python/ex07_aircraft.py`

[Notebook in Colab](#)

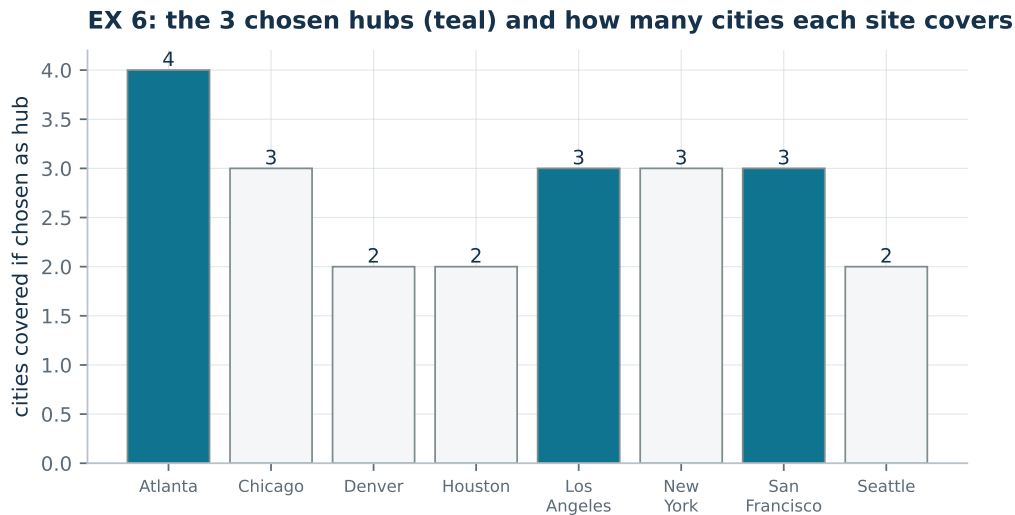


Figure 3: EX 6: for every city, how many it would cover as a hub; in teal the three chosen ones.

EX 7 — which orders to accept

A company builds custom-designed aircraft; before producing for a client a fixed set-up cost has to be paid. Three clients have ordered, respectively, 3, 2 and 5 aircraft. The production capacity is 1000 hours. The company decides which orders to accept and how many units to produce for each (possibly fewer than requested). Set-up costs, unit profits and hours per aircraft:

	client 1	client 2	client 3
set-up (millions)	3	2	0
unit profit (millions)	2	3	1
hours per aircraft	200	400	200

The goal is to maximise the total profit net of the set-up costs.

The model.

The data are $n = 3$ clients, with unit profit p_j , setup cost f_j , hours per aircraft h_j and maximum order M_j , plus the production capacity $H = 1000$ hours. With x_j aircraft produced for client j (integer) and y_j the indicator of an accepted order:

EX 7

$$\begin{aligned}
& \max \sum_{j=1}^n (p_j x_j - f_j y_j) \\
& \text{subject to} \quad \sum_{j=1}^n h_j x_j \leq H, \\
& \quad x_j - M_j y_j \leq 0, \quad \forall j \in \{1, 2, \dots, n\}, \\
& \quad x_j \in \mathbb{Z}_{\geq 0}, \quad \forall j \in \{1, 2, \dots, n\}, \\
& \quad y_j \in \{0, 1\}, \quad \forall j \in \{1, 2, \dots, n\}.
\end{aligned}$$

EX 7 on the instance — primal

$$\begin{aligned}
& \max \quad 2x_1 + 3x_2 + x_3 - 3y_1 - 2y_2 \\
& \text{subject to} \quad 200x_1 + 400x_2 + 200x_3 \leq 1000 \\
& \quad x_1 - 3y_1 \leq 0 \\
& \quad x_2 - 2y_2 \leq 0 \\
& \quad x_3 - 5y_3 \leq 0 \\
& \quad x_1, x_2, x_3 \in \mathbb{Z}_{\geq 0} \\
& \quad y_1, y_2, y_3 \in \{0, 1\}
\end{aligned}$$

Description. A single constraint per client does two things at once: it caps the quantity at the number of aircraft ordered and it charges the set-up. Client 3 has a zero set-up, so y_3 does not appear in the objective and equals 1 in every optimal solution with $x_3 > 0$.

Heuristic: the primal bound.

Constructive heuristic on the net profit per hour, computed on the whole order: $3/600$, $4/800$ and $5/1000$, that is $1/200$ for all three. With equal ratios one scans in the natural order: client 1 is accepted with 3 aircraft (600 hours, net profit 3), then client 2 with a single aircraft (400 hours, net profit $3 - 2 = 1$), and the hours run out.

$$z(MILP) \geq LB = 4.$$

Dual: the dual bound.

With $\alpha \geq 0$ on the capacity and $\beta_j \geq 0$ on the links:

$$\min 1000 \alpha \quad \text{s.t.} \quad h_j \alpha + \beta_j \geq p_j, \quad M_j \beta_j \leq f_j.$$

The recipe: $\bar{\beta}_j = f_j/M_j$, that is the set-up spread over the aircraft ordered (1, 1 and 0), whence $\alpha \geq (p_j - \bar{\beta}_j)/h_j$ for every j , that is $\alpha \geq 1/200$ in all three cases. With $\bar{\alpha} = 1/200$,

$$z(MILP) \leq UB = 1000 \cdot \frac{1}{200} = 5.$$

Results.

	LB	UB	$z(LP)$	$z(LP^+)$	$z(MILP)$
EX 7 aircraft (max)	4	5	5	5	5

The optimum is 5 and is attained by *three* different plans: five aircraft for client 3; two for client 2 and one for the third; three for client 1 and two for the third. All three use exactly the 1000 hours. The dual bound is exact, and in this case the certificate settles the problem on its own: there is nothing left for the solver to do. Without set-up costs the optimum would rise to 9; with 1400 hours available, to 7.

EX 8 — Seminars

Script: `python/ex08_seminars.py`[Notebook in Colab](#)

EX 8 — selection with exact cardinality and non-adjacency

A student must attend *exactly two* sessions of a summer school. There are three seminars, each offered in four time slots (9–10, 10–11, 11–12, 12–13); every seminar is attended at most once and two sessions cannot occupy the same slot. The preference score of each session is:

	9–10	10–11	11–12	12–13
Seminar 1	8	6	5	3
Seminar 2	7	9	4	6
Seminar 3	5	7	8	9

The student never wants *two consecutive hours* of class. The total preference is to be maximised.

The model.

The data are $n = 3$ seminars, $m = 4$ slots, $q = 2$ sessions to attend and the score p_{sk} . With $x_{sk} = 1$ if seminar s is attended in slot k ($nm = 12$ binaries):

EX 8

$$\begin{aligned}
& \max \sum_{s=1}^n \sum_{k=1}^m p_{sk} x_{sk} \\
& \text{subject to} \quad \sum_{s=1}^n x_{sk} \leq 1, & \forall k \in \{1, 2, \dots, m\}, \\
& \quad \sum_{k=1}^m x_{sk} \leq 1, & \forall s \in \{1, 2, \dots, n\}, \\
& \quad \sum_{s=1}^n \sum_{k=1}^m x_{sk} = q, \\
& \quad \sum_{s=1}^n (x_{sk} + x_{s,k+1}) \leq 1, & \forall k \in \{1, 2, \dots, m-1\}, \\
& \quad x_{sk} \in \{0, 1\}, & \forall s \in \{1, 2, \dots, n\}, \forall k \in \{1, 2, \dots, m\}.
\end{aligned}$$

EX 8 on the instance — primal

$$\begin{aligned}
& \max 8x_{11} + 6x_{12} + 5x_{13} + 3x_{14} + 7x_{21} + 9x_{22} + 4x_{23} + 6x_{24} + 5x_{31} + 7x_{32} + 8x_{33} + 9x_{34} \\
& \text{subject to} \quad x_{11} + x_{21} + x_{31} \leq 1 \\
& \quad x_{12} + x_{22} + x_{32} \leq 1 \\
& \quad x_{13} + x_{23} + x_{33} \leq 1 \\
& \quad x_{14} + x_{24} + x_{34} \leq 1 \\
& \quad x_{11} + x_{12} + x_{13} + x_{14} \leq 1 \\
& \quad x_{21} + x_{22} + x_{23} + x_{24} \leq 1 \\
& \quad x_{31} + x_{32} + x_{33} + x_{34} \leq 1 \\
& \quad x_{11} + x_{12} + x_{13} + x_{14} + x_{21} + x_{22} + x_{23} + x_{24} + x_{31} + x_{32} + x_{33} + x_{34} = 2 \\
& \quad x_{11} + x_{12} + x_{21} + x_{22} + x_{31} + x_{32} \leq 1 \\
& \quad x_{12} + x_{13} + x_{22} + x_{23} + x_{32} + x_{33} \leq 1 \\
& \quad x_{13} + x_{14} + x_{23} + x_{24} + x_{33} + x_{34} \leq 1 \\
& \quad x_{11}, x_{12}, x_{13}, x_{14}, x_{21}, x_{22}, x_{23}, x_{24}, x_{31}, x_{32}, x_{33}, x_{34} \in \{0, 1\}
\end{aligned}$$

Description. with $q = 2$: four slot constraints, three seminar constraints, one *exact* cardinality constraint and three non-adjacency constraints between consecutive slots.

Heuristic.

Constructive heuristic: take the feasible session with the highest score, then the best compatible one. Step 1: seminar 3 in slot 4 (score 9). Step 2: among the sessions that do not use seminar 3, do not occupy slot 4 and are not adjacent to it, the best is seminar 2 in slot 2 (score 9). In a **maximisation** the value is a *lower* bound: $LB = 18$.

Dual and upper bound.

With $\alpha_k \geq 0$ for the slots, $\beta_s \geq 0$ for the seminars, γ free for the exact cardinality and $\delta_k \geq 0$ for non-adjacency:

Dual of the LP relaxation of EX 8

$$\begin{aligned}
 \min \quad & \sum_{k=1}^m \alpha_k + \sum_{s=1}^n \beta_s + q\gamma + \sum_{k=1}^{m-1} \delta_k \\
 \text{subject to} \quad & \alpha_k + \beta_s + \gamma + \sum_{k': k \in \{k', k'+1\}} \delta_{k'} \geq p_{sk}, \quad \forall s \in \{1, 2, \dots, n\}, \forall k \in \{1, 2, \dots, m\}, \\
 & \alpha_k \geq 0, \quad \forall k \in \{1, 2, \dots, m\}, \\
 & \beta_s \geq 0, \quad \forall s \in \{1, 2, \dots, n\}, \\
 & \gamma \geq 0, \\
 & \delta_k \geq 0, \quad \forall k \in \{1, 2, \dots, m-1\}.
 \end{aligned}$$

Description. The variable α_k is the price of time slot k , β_s that of seminar s , γ the price of the cardinality constraint and δ_k that of the non-adjacency constraint between slots k and $k+1$. The objective adds up the right-hand sides of those constraints at those prices. The dual constraints are the columns of the x_{sk} : attending seminar s in slot k takes the slot, takes the seminar, consumes one unit of cardinality and touches the non-adjacency constraints containing k ; the total must cover the preference p_{sk} .

The shortest recipe: everything at zero except $\bar{\gamma} = \max_{s,k} p_{sk} = 9$. Every dual constraint becomes $9 \geq p_{sk}$, true by construction, and the value is $q\bar{\gamma} = 2 \cdot 9 = 18$. It reads: "two sessions are attended, and none is worth more than the best one".

Dual recipe	UB
everything at zero except $\gamma = \max_{s,k} p_{sk}$	18
$\gamma = 0, \alpha_k = \max_s p_{sk}$ ("every slot is worth at most its best")	34

The first is optimal for the relaxation; the second, apparently more natural, is almost twice as large. The recipe matters.

The bounds.

The optimum is $z(MILP) = 18$: seminar 2 in slot 2 and seminar 3 in slot 4 — with a break in between, as required.

LB (constructive heuristic)	$z(MILP)$	$z(LP)$	UB (dual by hand)	heuristic gap
18	18	18	18	0.0%

Here the two hand-built bounds *close* the problem: the heuristic is worth 18, the dual 18, so $z(MILP) = 18$ is proved without the solver.

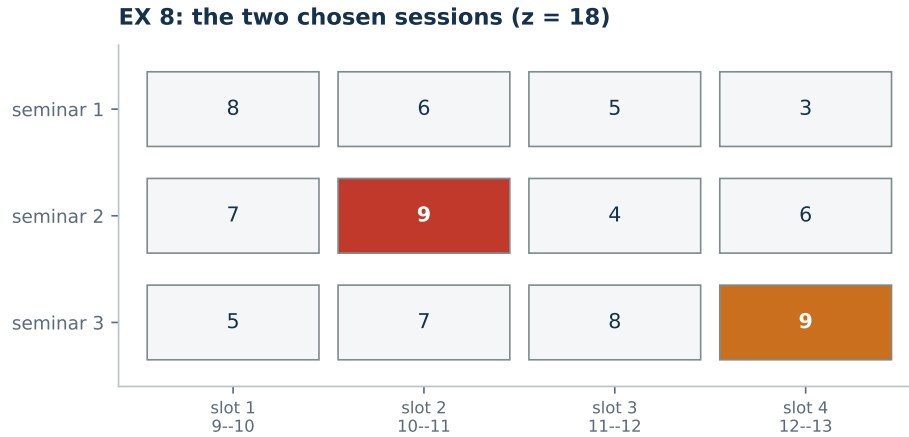


Figure 4: EX 8: the grid of preferences; in teal the two chosen sessions.

EX 9 — The eight queens

Script: `python/ex09_queens.py`

[Notebook in Colab](#)

EX 9 — the largest number of queens

On an 8×8 chessboard one wants to place the largest number of queens so that no queen can capture another in a single move. In chess a queen moves any number of squares along the same row, the same column or either diagonal.

The model.

The board is $n \times n$, with $n = 8$. With $x_{ij} = 1$ if there is a queen on square (i, j) ($n^2 = 64$ binaries):

EX 9

$$\begin{aligned}
 & \max \sum_{i=1}^n \sum_{j=1}^n x_{ij} \\
 & \text{subject to} \quad \sum_{j=1}^n x_{ij} \leq 1, & \forall i \in \{1, 2, \dots, n\}, \\
 & \quad \sum_{i=1}^n x_{ij} \leq 1, & \forall j \in \{1, 2, \dots, n\}, \\
 & \quad \sum_{i-j=k} x_{ij} \leq 1, & \forall k \in \{-(n-1), -(n-2), \dots, n-1\}, \\
 & \quad \sum_{i+j=k} x_{ij} \leq 1, & \forall k \in \{2, 3, \dots, 2n\}, \\
 & \quad x_{ij} \in \{0, 1\}, & \forall i \in \{1, 2, \dots, n\}, \forall j \in \{1, 2, \dots, n\}.
 \end{aligned}$$

Description. A set packing on four families of lines: 8 rows, 8 columns and 15+15 diagonals, 46 constraints in total. The indexing trick is all here: the "descending" diagonals are the sets with $i - j$ constant, the "ascending" ones the sets with $i + j$ constant.

Heuristic: the primal bound.

Constructive heuristic row by row: in row i one picks the first free column that is not attacked by the queens already placed; if there is none, the row stays empty. It never backtracks. On the instance the constructive heuristic places queens on $(1, 1)$, $(2, 3)$, $(3, 5)$, $(4, 2)$, $(5, 4)$ and then gets stuck: rows 6, 7 and 8 have no free squares left.

$$z(MILP) \geq LB = 5.$$

Dual: the dual bound.

The dual is a set covering:

$$\begin{aligned} \min \quad & \sum_{i=1}^8 \alpha_i + \sum_{j=1}^8 \beta_j + \sum_{k=-7}^7 \gamma_k + \sum_{k=2}^{16} \delta_k \\ \text{s.t.} \quad & \alpha_i + \beta_j + \gamma_{i-j} + \delta_{i+j} \geq 1, \quad \forall i \in \{1, 2, \dots, 8\}, \forall j \in \{1, 2, \dots, 8\}. \end{aligned}$$

For example one can use a single line: $\bar{\alpha}_i = 1$ for every row, everything else zero. Every square (i, j) has $\alpha_i = 1 \geq 1$, so the solution is feasible and is worth 8. It is the dual translation of the sentence "at most one queen sits in each row".

$$z(MILP) \leq UB = 8.$$

Results.

	LB	UB	$z(LP)$	$z(LP^+)$	$z(MILP)$
EX 9 queens (max)	5	8	8	8	8

The dual bound is attained: the solution found is optimal, and one knows it without trusting the solver. The constructive heuristic, on the other hand, stops at 5: it is the heuristic, not the bound, that leaves the 37.5% gap.

Two variants.

On $n \times n$ boards with $n = 4, 5, 6$ the optimum is always n , and the certificate $\bar{\alpha}_i = 1$ proves it. For $n = 2$ and $n = 3$, instead, the optimum is 1 and 2: the bound n stays valid but is not attainable, and it is the first case in which this dual recipe does not settle the problem. Removing the diagonal constraints one gets *rooks* instead of queens: the model becomes an assignment, the matrix is totally unimodular and the LP relaxation already gives an integer value.

EX 10 — Tools of a CNC machine

Script: `python/ex10_tools.py`

[Notebook in Colab](#)

EX 10 — selection with a tool set

A CNC machine can perform five operations; each one requires a subset of six tools and, if performed, yields a profit:

Operation	Required tools	Profit (€)
1	1, 3, 4, 5	2000
2	1, 2, 6	1500
3	1, 2, 3, 5	1800
4	2, 4, 5	1700
5	5, 6	800

The magazine of the machine holds at most $k = 4$ tools at a time. The configuration of maximum profit is wanted.

The model.

The data are $n = 5$ operations of profit p_i , $m = 6$ tools and a magazine holding $k = 4$ tools; T_i is the set of tools required by operation i . With $x_i = 1$ if operation i is performed (n binaries) and $y_j = 1$ if tool j is loaded (m binaries):

EX 10

$$\begin{aligned}
 & \max \sum_{i=1}^n p_i x_i \\
 & \text{subject to} \quad \sum_{j=1}^m y_j \leq k, \\
 & \quad x_i - y_j \leq 0, \quad \forall i \in \{1, 2, \dots, n\}, \forall j \in T_i, \\
 & \quad x_i \in \{0, 1\}, \quad \forall i \in \{1, 2, \dots, n\}, \\
 & \quad y_j \in \{0, 1\}, \quad \forall j \in \{1, 2, \dots, m\}.
 \end{aligned}$$

EX 10 on the instance — primal

$$\begin{array}{ll}
\max & 2000x_1 + 1500x_2 + 1800x_3 + 1700x_4 + 800x_5 \\
\text{subject to} & y_1 + y_2 + y_3 + y_4 + y_5 + y_6 \leq 4 \\
& x_1 - y_1 \leq 0 \\
& x_1 - y_3 \leq 0 \\
& x_1 - y_4 \leq 0 \\
& x_1 - y_5 \leq 0 \\
& x_2 - y_1 \leq 0 \\
& x_2 - y_2 \leq 0 \\
& x_2 - y_6 \leq 0 \\
& x_3 - y_1 \leq 0 \\
& x_3 - y_2 \leq 0 \\
& x_3 - y_3 \leq 0 \\
& x_3 - y_5 \leq 0 \\
& x_4 - y_2 \leq 0 \\
& x_4 - y_4 \leq 0 \\
& x_4 - y_5 \leq 0 \\
& x_5 - y_5 \leq 0 \\
& x_5 - y_6 \leq 0 \\
& x_1, x_2, x_3, x_4, x_5 \in \{0, 1\} \\
& y_1, y_2, y_3, y_4, y_5, y_6 \in \{0, 1\}
\end{array}$$

Description. One magazine constraint and $\sum_i |T_i| = 4 + 3 + 4 + 3 + 2 = 16$ link constraints. It is the *disaggregated* activation of Section “Activation: aggregated and disaggregated form” read the other way round: here the operation is the object and the tools are the resources, and the operation needs *all* of its tools — a conjunction in the consequent, which by the table of Chapter “Logic and binary variables” costs one constraint per tool.

Heuristic.

Constructive heuristic by decreasing profit: the operations are scanned from the most profitable and the tool set of each is loaded, if the magazine allows it.

- **Operation 1** (2000, tools 1, 3, 4, 5): 4 new tools are needed and the magazine reaches exactly 4: it runs.
- **Operations 3, 4, 2, 5**: each would need at least one more tool, and the magazine is full: they are discarded.

In a **maximisation** the value is a *lower* bound: $LB = 2000$.

Dual and upper bound.

Relaxing to $x_i, y_j \geq 0$, with $\alpha \geq 0$ for the magazine constraint and $\beta_{ij} \geq 0$ for each link constraint:

Dual of the LP relaxation of EX 10

$$\begin{aligned}
& \min \quad k \alpha \\
& \text{subject to} \quad \sum_{j \in T_i} \beta_{ij} \geq p_i, & \forall i \in \{1, 2, \dots, n\}, \\
& \quad \sum_{i: j \in T_i} \beta_{ij} \leq \alpha, & \forall j \in \{1, 2, \dots, m\}, \\
& \quad \alpha \geq 0, \\
& \quad \beta_{ij} \geq 0, & \forall i \in \{1, 2, \dots, n\}, \forall j \in T_i.
\end{aligned}$$

Description. It reads like this: β_{ij} is the share of the profit of operation i attributed to tool j ; the first group asks the shares to cover the whole profit, the second that no tool receives more than α ; the objective pays α for each of the k slots in the magazine.

The recipe. The profit of each operation is spread in equal parts over the tools it needs, $\bar{\beta}_{ij} = p_i/|T_i|$: the first group is satisfied with equality by construction. Then $\bar{\alpha} = \max_j \sum_{i: j \in T_i} \bar{\beta}_{ij}$, the load of the most demanded tool.

$\bar{\beta}_{ij} = p_i/ T_i $	load of the tool					
	1	2	3	4	5	6
op. 1: $2000/4 = 500$	500		500	500	500	
op. 2: $1500/3 = 500$	500	500				500
op. 3: $1800/4 = 450$	450	450	450		450	
op. 4: $1700/3 = 1700/3$		1700/3		1700/3	1700/3	
op. 5: $800/2 = 400$					400	400
total	1450	4550/3	950	3200/3	5750/3	900

The critical tool is number 5, required by four operations out of five: hence $\bar{\alpha} = 5750/3$ and

$$UB = k \bar{\alpha} = 4 \cdot \frac{5750}{3} = \frac{23000}{3} \approx 7666.7.$$

A choice that comes naturally is $\bar{\alpha} = 2000$ with all $\bar{\beta}_{ij} = 900$: the constraints on the operations are satisfied ($2 \cdot 900 = 1800 \geq 800$ is the tightest case) and the value would be $4 \cdot 2000 = 8000$. **But that solution is not feasible.** The second group of constraints asks $\sum_{i: j \in T_i} \beta_{ij} \leq \alpha$, and tool 5 serves *four* operations: it would receive $4 \cdot 900 = 3600 > 2000$. The violation is 1600. The number 8000 is not a bound: it does not come from any feasible dual solution.

This is why the course protocol does not ask to *propose* a dual solution but to *check* it: the script verifies every constraint, and an **assert** fails if the violation exceeds the tolerance. With the correct recipe the bound is $23000/3 \approx 7666.7$, that is, *better* than the one hoped for by getting it wrong.

The bounds.

$z(MILP) = 2500$: tools 2, 4, 5, 6 are loaded and operations 4 (1700) and 5 (800) are performed. No four-tool configuration yields more: with $\{1, 3, 4, 5\}$ only operation 1 runs (2000), with $\{1, 2, 3, 5\}$ only operation 3 (1800).

LB (constructive heuristic)	$z(MILP)$	$z(LP)$	$z(LP^+)$	UB (dual by hand)	heuristic gap
2000	2500	5200	5200	23000/3	20.0%

Why the relaxation is so weak here

$z(LP) = 5200$ against $z(MILP) = 2500$: more than double. In the continuous problem one can load "a bit" of every tool — for instance $y_j = 2/3$ for six tools, which respects $\sum_j y_j \leq 4$ — and run a fraction of *every* operation. The integrality of the magazine is everything: this is the case where hand-built bounds, however well constructed, are not enough and the work has to be done by the solver. One way to tighten is to add the valid inequalities $\sum_{j \in T_i} y_j \geq |T_i|x_i$ (the aggregated form of the links), or to observe that any operation with $|T_i| > k$ can be fixed to zero.

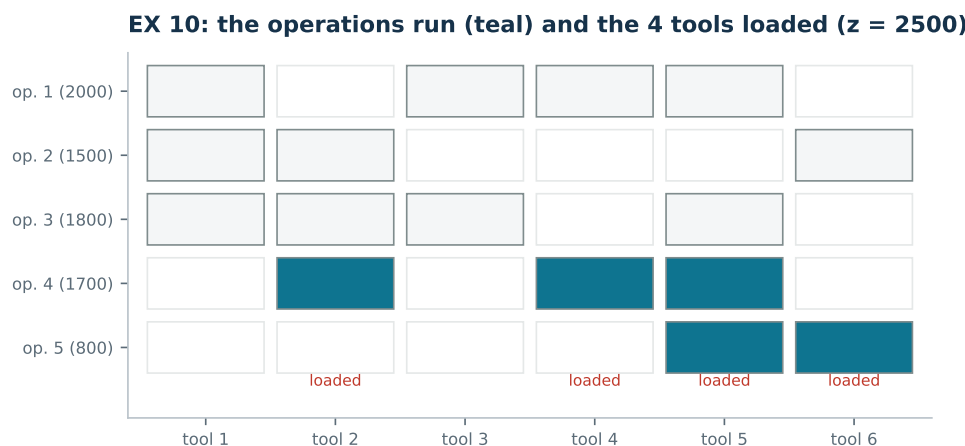


Figure 5: EX 10: the operation-tool matrix. In teal the operations performed; the annotated columns are the four loaded tools.

EX 11 — Balancing between two workers

Script: `python/ex11_balancing.py`

[Notebook in Colab](#)

EX 11 — workloads as balanced as possible

Four indivisible jobs of duration 2, 3, 6 and 7 must be assigned to two workers so that their workloads are as balanced as possible.

"Balanced workloads" is not an objective: it is a word. The two linear translations are those of Section "Min-max, max-min and the range": minimise the *maximum* of the loads,

or minimise their *difference*. Since the total $q = 18$ is constant, the identity

$$\max(w_1, w_2) = \frac{q}{2} + \frac{|w_1 - w_2|}{2}$$

holds, so the two models have the *same* optimal solutions — but different optimal values: 9 for the first, 0 for the second. Reporting "9" as the "difference between the loads" is a mistake, and so is reporting "0" as the "maximum load".

The model (min-max).

The data are $n = 4$ jobs of duration d_j and the total load $q = \sum_{j=1}^n d_j = 18$. With $x_j = 1$ if job j goes to worker 1 and $z \geq 0$ the maximum load:

EX 11, min-max version

$$\begin{aligned} & \min \quad z \\ & \text{subject to} \quad \sum_{j=1}^n d_j x_j - z \leq 0, \\ & \quad \quad \quad - \sum_{j=1}^n d_j x_j - z \leq -q, \quad (\text{load of worker 2, equal to } q - \sum_j d_j x_j) \\ & \quad \quad \quad x_j \in \{0, 1\}, \quad \forall j \in \{1, 2, \dots, n\}, \\ & \quad \quad \quad z \geq 0. \end{aligned}$$

EX 11 on the instance — primal

$$\begin{aligned} & \min \quad z \\ & \text{subject to} \quad 2x_1 + 3x_2 + 6x_3 + 7x_4 - z \leq 0 \\ & \quad \quad \quad -2x_1 - 3x_2 - 6x_3 - 7x_4 - z \leq -18 \\ & \quad \quad \quad x_1, \quad x_2, \quad x_3, \quad x_4 \in \{0, 1\} \\ & \quad \quad \quad z \geq 0 \end{aligned}$$

Description. The objective is the load of the busier worker. The first constraint says that z is no smaller than the load of worker 1; the second that it is no smaller than that of worker 2, which equals $q - \sum_j d_j x_j$. Together they impose $z \geq \max(w_1, w_2)$, and the minimisation turns the inequality into an equality on the busier of the two.

Heuristic.

LPT (Chapter "Constructive heuristics"): the jobs in decreasing order of duration, each to the least loaded worker. Job 4 (7) to worker 1; job 3 (6) to worker 2; job 2 (3) to worker 2, which reaches 9; job 1 (2) to worker 1, which reaches 9. Loads (9, 9), so $UB = 9$.

Dual and lower bound.

Relaxing to $x_j \geq 0$ and using the conversion table — a *minimisation* primal with $\leq$ constraints, hence dual variables $\pi_1, \pi_2 \leq 0$:

Dual of the LP relaxation of EX 11

$$\begin{aligned}
 & \max \quad -q \pi_2 \\
 & \text{subject to} \quad d_j (\pi_1 - \pi_2) \leq 0, & \forall j \in \{1, 2, \dots, n\}, \\
 & \quad \quad \quad -\pi_1 - \pi_2 \leq 1, & \text{(from the column of } z) \\
 & \quad \quad \quad \pi_1, \pi_2 \leq 0.
 \end{aligned}$$

Description. The variables π_1 and π_2 are the (non-positive) prices of the two min-max constraints. The objective prices the total load q at π_2 . The first group of constraints are the columns of the x_j : moving job j from one worker to the other shifts both loads by d_j , and the balance cannot be positive because the x_j do not appear in the objective. The last constraint is the column of z , which appears in both constraints with coefficient -1 and in the primal objective with cost 1.

Since $d_j > 0$, the first constraints say $\pi_1 \leq \pi_2$. Setting $\pi_1 = \pi_2 = t$, the constraint from z gives $-2t \leq 1$, that is $t \geq -1/2$; the objective $-qt$ is largest at $t = -1/2$ and equals $LB = q/2 = 9$. Meaning: "the two loads add up to q , so the larger one is at least $q/2$ ".

The same thing with non-negative dual variables

Many treatments prefer dual variables ≥ 0 . It suffices to set $\alpha = -\pi_1$ and $\beta = -\pi_2$: the dual becomes $\max q\beta$ with $\alpha \leq \beta$ and $\alpha + \beta \leq 1$, and $\alpha = \beta = 1/2$ gives 9 again. It is the *same* solution written with the other sign; what one cannot do is declare variables ≥ 0 and then use one of them negative.

The bounds.

$z(MILP) = 9$ with jobs $\{1, 4\}$ to the first worker and $\{2, 3\}$ to the second: loads $(9, 9)$, perfectly balanced.

UB (LPT)	LB (dual by hand)	$z(LP)$	$z(MILP)$	heuristic gap
9	9	9	9	0.0%

All the numbers coincide: here the two hand-built bounds meet, and optimality is proved without solving the MILP. This happens because q is even and the jobs can be split exactly in half; with q odd the bound would become $\lceil q/2 \rceil$ and one more argument would be needed.

EX 12 — Shoes with a minimum production threshold

Script: `python/ex12_shoes_threshold.py`

[Notebook in Colab](#)

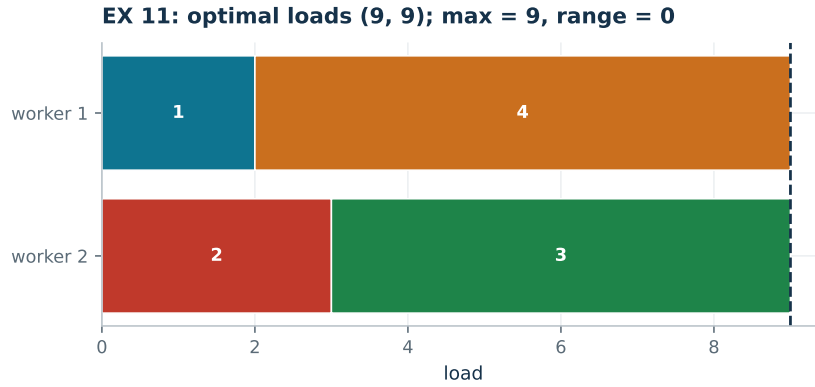


Figure 6: EX 11: the optimal loads. The dashed line is $q/2$, the dual bound.

EX 12 — three types of shoe with a threshold

A shoe company produces hiking boots, loafers and walking shoes. The table reports the resources needed for one pair, the minimum quantity to be produced if the type is produced, and the selling price:

Type	leather (g)	hours	nails	minimum	price
hiking boots	850	3	20	100	150
loafers	600	2	15	200	120
walking shoes	700	2.5	20	150	130

There are 120 000 grams of leather, 7000 machine hours and 40 000 nails available. The goal is to maximise the revenue.

The model.

It is again model 9.3 without the bonus, with three resources. The big-Ms read off the data are $M = (141, 200, 171)$, all limited by the leather. For the loafers threshold and maximum coincide (200): the variable x_2 is in fact a binary multiplied by 200.

Heuristic: the primal bound.

Constructive heuristic on the price per gram of leather: $3/17$ for the boots, $1/5$ for the loafers, $13/70$ for the walking shoes. One starts from the loafers, produces 200 of them and the leather runs out exactly. No other type reaches its own threshold.

$$z(MILP) \geq LB = 24\,000.$$

Dual: the dual bound.

With $\ell = \beta = 0$ and a single resource priced:

$$\text{leather: } \frac{1}{5} \Rightarrow 24\,000, \quad \text{hours: } 60 \Rightarrow 420\,000, \quad \text{nails: } 8 \Rightarrow 320\,000.$$

The better bound is the one from the leather:

$$z(MILP) \leq UB = 24\,000.$$

Results.

	LB	UB	$z(LP)$	$z(LP^+)$	$z(MILP)$
EX 12 shoes with threshold (max)	24 000	24 000	24 000	24 000	24 000

All the numbers coincide: the sandwich closes immediately. The reason is that a single resource is tight, the leather, and the type that gets the most out of it per gram uses it up on its own, staying within its own threshold and its own maximum. It is a didactically useful case precisely because it is easy: it shows that when the dual bound matches a feasible solution, the problem is over.

Two variations change the picture. Doubling the thresholds, the one for the loafers becomes 400 pairs while the leather allows at most 200: no type reaches its own threshold any more and production stops altogether, $z(MILP) = 0$. Raising the leather to 200 000 grams, 333 loafers are produced for 39 960 euros, and the big-Ms must be recomputed: they become (235, 333, 285). Big-M values are not constants of the model, but functions of the data of the instance.

EX 13 — Mutual funds bought in lotsScript: `python/ex13_funds.py`[Notebook in Colab](#)**EX 13 — two mutual funds**

After a lottery win, an investor must decide how to allocate 100 million lire. A friend suggests two types of mutual fund. Shares can only be bought in indivisible lots: one lot of the first type costs 12 million, one of the second 20 million. The estimated annual return is $1/6$ of the capital invested for the first type and 15% for the second. Moreover the number of lots of the second type cannot exceed 50% of the total number of lots bought. The goal is to maximise the expected annual return.

The model.

The return of one lot is $12 \cdot 1/6 = 2$ million for the first fund and $20 \cdot 0.15 = 3$ for the second. The share constraint $x_2 \leq \frac{1}{2}(x_1 + x_2)$, multiplied by 2 and moved to the left, becomes $-x_1 + x_2 \leq 0$. With x_j the number of lots of fund j :

EX 13 — the model written out in full

$$\begin{aligned}
 &\max \quad 2x_1 + 3x_2 \\
 &\text{subject to} \quad 12x_1 + 20x_2 \leq 100 \\
 &\quad \quad \quad -x_1 + x_2 \leq 0 \\
 &\quad \quad \quad x_1, \quad x_2 \in \mathbb{Z}_{\geq 0}
 \end{aligned}$$

Description. An integer, non-binary knapsack: the variables count lots, not choices.

Heuristic: the primal bound.

Constructive heuristic on the return per million invested: $2/12 = 1/6$ for the first fund and $3/20$ for the second. One starts from the first, buys 8 lots (96 million) and is left with 4 million,

not enough for another lot of either type.

$$z(MILP) \geq LB = 16.$$

Dual: the dual bound.

With $\alpha \geq 0$ on the budget and $\beta \geq 0$ on the share:

$$\min 100\alpha \quad \text{s.t.} \quad 12\alpha - \beta \geq 2, \quad 20\alpha + \beta \geq 3.$$

The original draft of this exercise proposed $\bar{\alpha} = 5/32$, $\bar{\beta} = 1/8$. Let us check the first constraint:

$$12 \cdot \frac{5}{32} - \frac{1}{8} = \frac{60}{32} - \frac{4}{32} = \frac{56}{32} = \frac{7}{4} < 2.$$

It is violated by $1/4$: that pair is *not* a feasible dual solution, and the value $100 \cdot 5/32$ is not a bound. A dual value may be read as a bound only after checking *all* the constraints, one by one.

The correct recipe is $\bar{\beta} = 0$ and

$$\bar{\alpha} = \max_j \frac{p_j}{c_j} = \max\left(\frac{2}{12}, \frac{3}{20}\right) = \frac{1}{6},$$

that is, a million is worth what it returns in the better fund. Both constraints become $c_j \bar{\alpha} \geq p_j$ and are satisfied, and

$$z(MILP) \leq UB = 100 \cdot \frac{1}{6} = \frac{50}{3} \approx 16.67.$$

Results.

	LB	UB	$z(LP)$	$z(LP^+)$	$z(MILP)$
EX 13 funds (max)	16	50/3	50/3	50/3	16

The heuristic finds the optimum, 8 lots of the first fund and none of the second, for a return of 16 million. The difference $50/3 - 16 = 2/3$ is the price of integrality: the relaxation would buy $25/3$ lots of the first fund, which cannot be bought in pieces.

On the data of the instance the share constraint does not bite, because the first fund returns more per million invested and at the optimum the second one is not bought at all. The share becomes active as soon as the second fund returns 20%, that is 4 million a lot: then without the share the optimum would be 20 (five lots of the second fund) and with the share it drops to 18 (three and three). It is the only way to see what a constraint really does: change the data until it bites.

EX 14 — Emergency department shifts

Script: `python/ex14_shifts.py`

[Notebook in Colab](#)

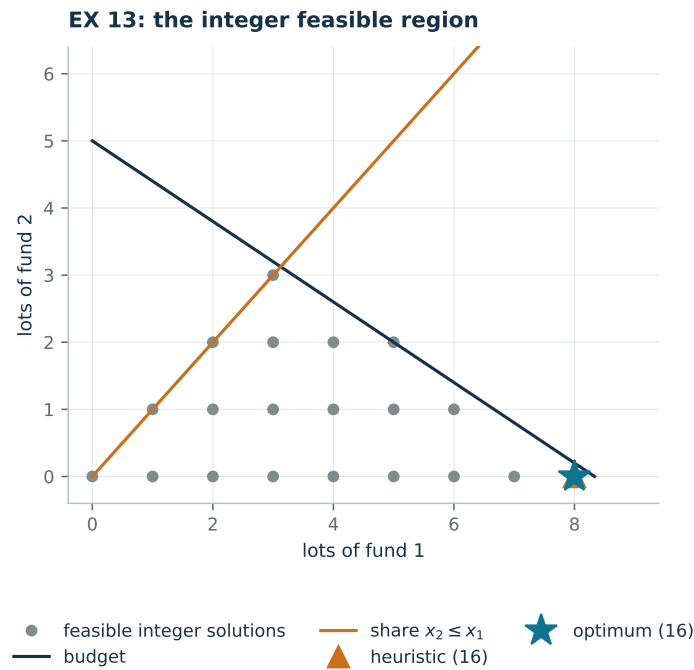


Figure 7: EX 13: the integer feasible region, with the budget line, the share line, the heuristic solution and the optimum (coinciding).

EX 14 — weekly shifts

In an emergency department the number of full-time staff members required is given for every day of the week:

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
staff	10	8	12	9	9	7	8

One full-time member can be replaced by two part-time members. The shift pattern is fixed: four consecutive full-time days, then one part-time day, then two rest days. The weekly cost of a worker depends on the days worked: 100 euros for every weekday (Monday to Friday), 110 for Saturday and 130 for Sunday; these costs are halved on part-time days. The goal is the weekly plan of minimum cost.

The model.

There are $n = 7$ shift patterns, one per starting day, and $m = 7$ days to cover, with requirement b_i . With x_j the number of workers following pattern j (integer variables), c_j the weekly cost of pattern j and a_{ij} the share of day i covered by pattern j (1 on the four full days, 1/2 on the part-time day, 0 on the two rest days):

EX 14

$$\begin{aligned}
& \min \sum_{j=1}^n c_j x_j \\
& \text{subject to} \quad \sum_{j=1}^n a_{ij} x_j \geq b_i, & \forall i \in \{1, 2, \dots, m\}, \\
& & x_j \in \mathbb{Z}_{\geq 0}, & \forall j \in \{1, 2, \dots, n\}.
\end{aligned}$$

EX 14 on the instance — primal

$$\begin{aligned}
& \min 450x_1 + 455x_2 + 475x_3 + 490x_4 + 490x_5 + 490x_6 + 480x_7 \\
& \text{subject to} \quad x_1 + \frac{1}{2}x_4 + x_5 + x_6 + x_7 \geq 10 \\
& \quad x_1 + x_2 + \frac{1}{2}x_5 + x_6 + x_7 \geq 8 \\
& \quad x_1 + x_2 + x_3 + \frac{1}{2}x_6 + x_7 \geq 12 \\
& \quad x_1 + x_2 + x_3 + x_4 + \frac{1}{2}x_7 \geq 9 \\
& \quad \frac{1}{2}x_1 + x_2 + x_3 + x_4 + x_5 \geq 9 \\
& \quad \frac{1}{2}x_2 + x_3 + x_4 + x_5 + x_6 \geq 7 \\
& \quad \frac{1}{2}x_3 + x_4 + x_5 + x_6 + x_7 \geq 8 \\
& \quad x_1, x_2, x_3, x_4, x_5, x_6, x_7 \in \mathbb{Z}_{\geq 0}
\end{aligned}$$

Description. It is a *set covering* with coefficients 1 and 1/2 and integer, non-binary variables. The costs are computed from the data:

Pattern	full days	part-time	cost
1 (starts Mon)	Mon, Tue, Wed, Thu	Fri	450
2 (starts Tue)	Tue, Wed, Thu, Fri	Sat	455
3 (starts Wed)	Wed, Thu, Fri, Sat	Sun	475
4 (starts Thu)	Thu, Fri, Sat, Sun	Mon	490
5 (starts Fri)	Fri, Sat, Sun, Mon	Tue	490
6 (starts Sat)	Sat, Sun, Mon, Tue	Wed	490
7 (starts Sun)	Sun, Mon, Tue, Wed	Thu	480

For instance pattern 3 costs $100 + 100 + 100 + 110$ for the four full days (Wed, Thu, Fri, Sat) plus $130/2 = 65$ for the part-time Sunday, that is 475 in total.

Heuristic: the primal bound.

Constructive covering heuristic: while some demand is uncovered one adds a copy of the pattern with the lowest ratio

$$\frac{\text{cost}}{\text{demand actually covered}}.$$

The ratio is recomputed at every step, because the residual demand changes. On the instance the constructive heuristic adds 18 shifts: eight of pattern 1 (at 100 euros per unit of demand covered), then four of pattern 3, two of pattern 5, one of pattern 4 and three of pattern 7, with steadily worsening ratios, up to 480 for the last unit.

$$z(\text{MILP}) \leq UB = 8410.$$

Dual: the dual bound.

The dual is

$$\max \sum_{i=1}^7 b_i \pi_i \quad \text{s.t.} \quad \sum_{i=1}^7 a_{ij} \pi_i \leq c_j, \quad \forall j \in \{1, 2, \dots, 7\}, \quad \pi \geq 0.$$

For example one can use the *best ratio* one: put the same price t on all the days. Every pattern covers $4 + 1/2 = 9/2$ person-days, so the dual constraint becomes $\frac{9}{2} t \leq c_j$ for every pattern, and the largest feasible value is

$$t = \min_j \frac{c_j}{9/2} = \min \left(100, \frac{910}{9}, \frac{950}{9}, \frac{980}{9}, \frac{980}{9}, \frac{980}{9}, \frac{320}{3} \right) = 100,$$

attained by pattern 1. The total demand is $10 + 8 + 12 + 9 + 9 + 7 + 8 = 63$, so

$$z(MILP) \geq LB = 100 \cdot 63 = 6300.$$

Results.

	UB	LB	$z(LP)$	$z(LP^+)$	$z(MILP)$
EX 14 shifts	8 410	6 300	115970/17	115970/17	7 060

The optimal plan hires 5 workers on pattern 1, 4 on pattern 3, 1 on pattern 4, 2 on pattern 5 and 3 on pattern 7, at a cost of 7060. The coverage is saturated on Wednesday, Saturday and Sunday, and in surplus on the other days. The heuristic gap is 19.1%; the difference $z(MILP) - z(LP) = 7060 - 115970/17 \approx 238$ is the price of integrality, that is of the fact that people are hired one at a time.

Is the part-time day worth it?

Let us compare the contract of the statement with an alternative one: four full days and *three* rest days, with no part-time. The minimum price per day covered is the same in both cases, 100 euros. And yet the optimum of the second contract is 6710, against the 7060 of the first: the part-time day is worth less. The reason is that the half day falls on a fixed day and often ends up where the coverage is already there; the flexibility lost costs more than the extra coverage is worth. It is an example in which the conclusion "more coverage at the same unit price is always better" is false, and only the model shows it.

EX 15 — The music school timetable

Script: `python/ex15_timetable.py`

[Notebook in Colab](#)

EX 15 — weekly timetable

A music school needs to build its weekly timetable. There are four afternoons (Monday, Tuesday, Wednesday and Thursday), each of three hours. 6 hours of guitar, 3 of violin, 2 of piano and 1 of harp must be placed. Every day must include lessons of at least two different instruments. Piano and harp lessons cannot be in consecutive hours of the same day, because of a conflict between the two teachers. Every teacher has preferences to be avoided, if possible: the guitar teacher prefers not to teach at hour 1 on Monday and Tuesday, the

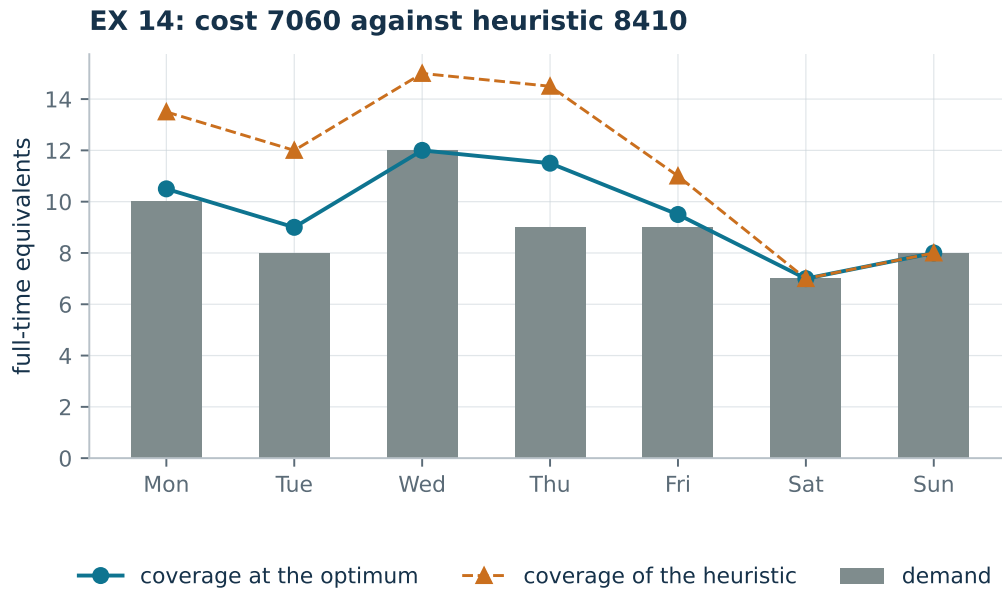


Figure 8: EX 14: daily demand and coverage, in the heuristic solution and in the optimal one.

violin teacher at hour 2 on Wednesday and Thursday, the piano teacher at hour 3 on any day, the harp teacher on Tuesday. The goal is to minimise the number of hours that violate a preference.

The model.

With D the days, T the hours, I the instruments, $x_{dti} = 1$ if at hour t of day d instrument i is taught, and $y_{di} = 1$ if instrument i appears on day d :

EX 15

$$\begin{aligned}
& \min \sum_{d \in D} \sum_{t \in T} \sum_{i \in I} c_{dti} x_{dti} \\
& \text{subject to} \quad \sum_{d \in D} \sum_{t \in T} x_{dti} = h_i, & \forall i \in I, \\
& \quad \sum_{i \in I} x_{dti} \leq 1, & \forall d \in D, \forall t \in T, \\
& \quad \sum_{i \in I} y_{di} \geq 2, & \forall d \in D, \\
& \quad x_{dti} - y_{di} \leq 0, & \forall d \in D, \forall t \in T, \forall i \in I, \\
& \quad y_{di} - \sum_{t \in T} x_{dti} \leq 0, & \forall d \in D, \forall i \in I, \\
& \quad x_{dt3} + x_{d,t+1,4} \leq 1, & \forall d \in D, \forall t \in \{1, 2\}, \\
& \quad x_{dt4} + x_{d,t+1,3} \leq 1, & \forall d \in D, \forall t \in \{1, 2\}, \\
& \quad x_{dti} \in \{0, 1\}, & \forall d \in D, \forall t \in T, \forall i \in I, \\
& \quad y_{di} \in \{0, 1\}, & \forall d \in D, \forall i \in I.
\end{aligned}$$

Description. The hours to place are $6 + 3 + 2 + 1 = 12$ and the slots are $4 \cdot 3 = 12$: the two figures coincide, so every slot holds exactly one lesson.

The original draft of this exercise contained only the link $x_{dti} \leq y_{di}$, and not its converse. With that direction alone the constraint $\sum_i y_{di} \geq 2$ is *always* satisfied: it is enough to set $y_{di} = 1$ without teaching. Solving the model written that way, the solver returns a timetable in which Thursday has a single instrument — three hours of guitar — and yet all the constraints hold.

The link $x_{dti} \leq y_{di}$ says "if there is a lesson then the indicator is on", not the converse. One also needs

$$y_{di} \leq \sum_{t \in T} x_{dti},$$

that is technique 3.10 (if and only if). It is the same mistake, and the same correction, as in problem 10.3: when a constraint *counts* indicators, the link must hold in both directions.

A feasible solution.

With the corrected model one builds by hand a timetable that violates no preference:

	hour 1	hour 2	hour 3
Monday	piano	guitar	guitar
Tuesday	violin	guitar	guitar
Wednesday	piano	guitar	guitar
Thursday	violin	harp	violin

One checks: 6 hours of guitar, 3 of violin, 2 of piano, 1 of harp; at least two instruments a day; piano and harp never on the same day, hence never consecutive; no preference violated.

$$z(MILP) \leq UB = 0.$$

The lower bound.

All the costs c_{dti} are 0 or 1, so the objective is a sum of non-negative terms and $z(MILP) \geq 0$ with no dual needed. The hand-built timetable is worth 0: it is optimal.

$$LB = z(LP) = z(LP^+) = z(MILP) = UB = 0.$$

The model therefore serves to *prove* that a conflict-free timetable exists, not to find a less bad one.

What happens if the preferences get tighter.

Two variants make the problem non-trivial.

- If the guitar teacher prefers not to teach at hour 1 nor at hour 2 of any day, only 4 slots stay penalty-free for the guitar (the hour 3 of the four days), but the hours to place are 6: at least $6 - 4 = 2$ lessons will violate the preference. It is a lower bound read off the data alone, and the solver confirms $z(MILP) = 2$: the count is exact.
- If at least *three* different instruments a day are required, the guitar can take at most one slot a day, that is 4 in total, while the guitar hours are 6. The model is infeasible, and one proves it by counting: Gurobi returns the status **INFEASIBLE**, as expected.

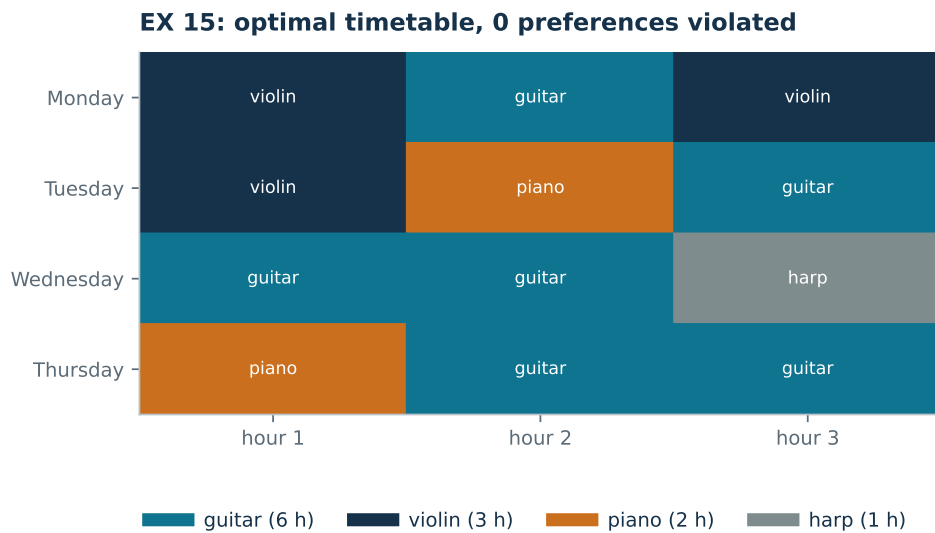


Figure 9: EX 15: the optimal timetable found by the solver, one colour per instrument.

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